Practical - 8

Practical Title: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition: Design the database for StudentHub with students, events and registrations. Create a PHP database connection using MySQL statements.

Theory :

1. Is the schema normalized properly?

Yes. The database has three tables: students, events, and registrations. Each table stores one kind of data in single-valued columns, and every non-key column depends only on the primary key of its table. Student details and event details are stored only once. The many-to-many relation between students and events is handled by the registrations table, so a student can join many events and an event can have many students without repeating any data.

2. Are primary keys and foreign keys correctly defined?

Yes. Every table has an AUTO_INCREMENT primary key: student_id, event_id, and registration_id. In registrations, student_id is a foreign key referencing students(student_id) and event_id is a foreign key referencing events(event_id), so a registration cannot exist for a missing student or event. The email column in students is UNIQUE, and UNIQUE (student_id, event_id) prevents the same student from registering twice for the same event.

3. Is the database connection handled securely?

Yes. The connection is created with PDO in db.php using the utf8mb4 charset, and the error mode is set to exceptions inside a try/catch block. All queries in index.php use prepared statements with named parameters (:name, :email, :course, :year), which protects against SQL injection. Input is trimmed and validated (email filter, year range 1 to 4), and output is escaped with htmlspecialchars() to prevent XSS. For a live website, the raw error message should be logged instead of displayed, and a separate MySQL user with limited rights should be used instead of root.

Implementation/ Output/Result

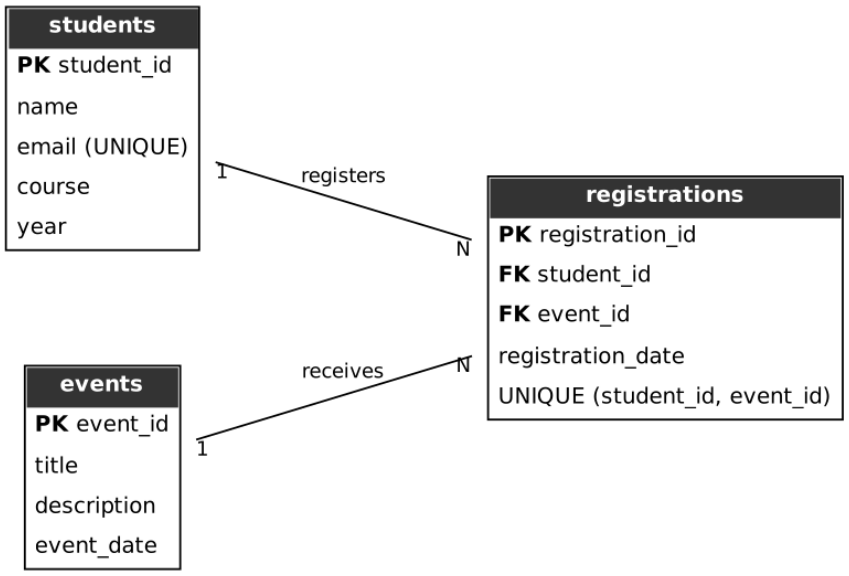
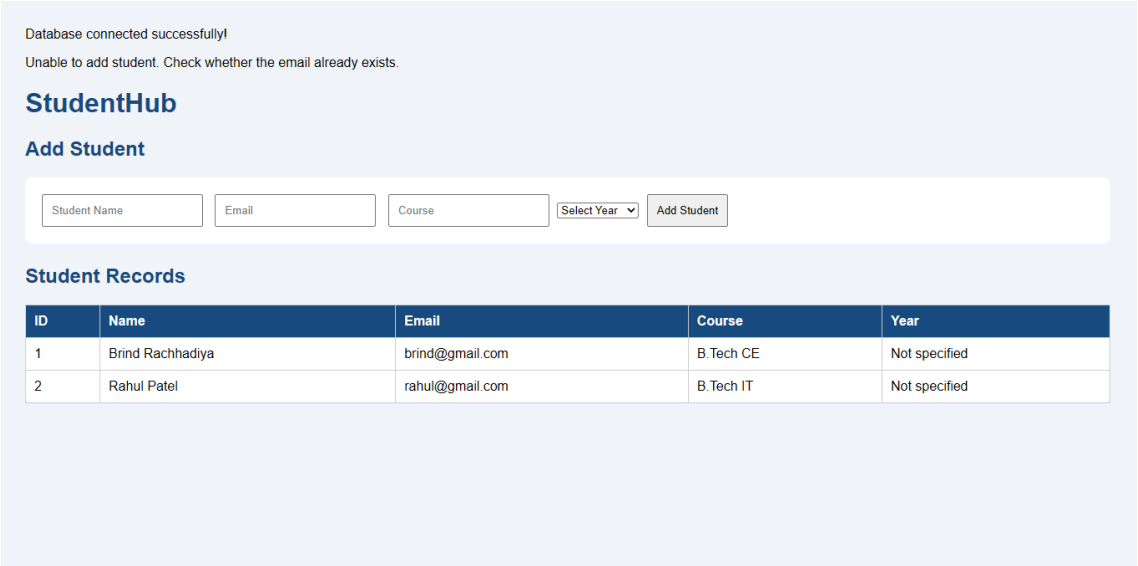


Figure 1: ER diagram of the StudentHub database (students, events, registrations)

OUTPUT:



GitHub Repository Link:

https://github.com/rachhadiyabrind-del/WDF_B/tree/main/Practical%208

Learning Outcomes:

- Design a normalized database schema using an ER model
- Define primary keys, foreign keys, and constraints in MySQL
- Create and import the database using phpMyAdmin
- Connect PHP to MySQL securely using PDO

- Use prepared statements to prevent SQL injection
- Handle database connection errors using try/catch

Faculty Remarks: (Keep this blank for faculty)